Trusted Aunti

A safety-aware assistant for adolescent girls in Kenya: contraception, sexual health, staying safe, and getting to a service.

Build report — the refined version.

github.com/George-techie/girl-effect-srh-rag · 176 tests · 8 governed sources, 1,693 chunks · 8 verified services across 8 routes

What the brief asked for

The previous submission was assessed as *overengineered in places*, with a request to see what a more refined version focused on delivery of the use case would look like. The scope also narrowed: **sexual and reproductive health, centred on contraception and family planning**, plus the empowerment outcomes that follow. Mental health and menstruation left the topic list.

This build is the answer to both. Same solution philosophy. Every component either justified by a measurement or deleted. **Refined does not mean smaller** — two components were added during the work, each because something measurable failed without them.

$0.0109 COST PER TURN	**52/52** ROUTING ACCURACY	**1.000** SAFEGUARDING RECALL & PRECISION	**5.8%** UNUSABLE, FROM 23.5%

Performance against the previous build

All previous figures below are read from that project's own recorded evaluation runs of 4 August 2026, 51 cases each, not reconstructed.

CONFIGURATION	ACC.	UNSAFE	UNHELPFUL	CALLS/TURN	TOKENS/TURN	COST/TURN	MEDIAN LATENCY
A — no safeguarding layer	0.784	7	1	0.98	3,938	–	5,476 ms
B — safeguarding, no evidence judge	0.882	0	5	2.65	3,848	–	7,058 ms
C — B plus the LLM evidence judge	0.745	0	12	3.53	6,838	–	7,030 ms
B+ — full profile, five model roles	0.843	1	6	3.78	8,599	$0.0189	12,410 ms
This build	52/52 †	–	3	0.69	4,946	$0.0109	5,200 ms

† Not the same measure. The previous action accuracy was scored by an LLM judge over 51 cases against a different scope and corpus; this build's 52/52 is deterministic routing accuracy. The columns that are directly comparable are calls, tokens, cost and latency — all measured over 51 and 52 cases respectively, priced with the same OpenRouter table.

On the refusal column: the previous build's "unhelpful actions" were scored by an LLM judge; this build's 3 are counted deterministically — blocked by the validator, no evidence found, or a provider error. Not the same instrument, so read it as the same order of magnitude rather than a precise ranking. The eight out-of-scope declines are excluded, because declining a question the service deliberately does not cover is correct behaviour rather than a refusal.

5.5× FEWER MODEL CALLS PER TURN	**1.7×** FEWER TOKENS PER TURN	**42%** LOWER COST PER TURN	**38%** OF TURNS NEED NO MODEL

What that comparison does and does not support

Cost and consumption are the robust numbers. They are properties of the architecture and do not move with network conditions. **0.69 model calls per turn against 3.78** — a 5.5× reduction — and **4,946 tokens per turn against 8,599**, which prices at **$0.0109 per turn against $0.0189**. Twenty of the fifty-two messages reached no model at all, and those are the turns where it matters most: every safeguarding reply is assembled from approved text and verified table rows in **0 ms**.

Per-turn call rate varies with the mix of questions. On the 52-message decision set, which is deliberately heavy in safeguarding and out-of-scope cases, it is 0.69. On the 38-turn conversation set, which is mostly questions, it is 1.03. Both are reported rather than the flattering one.

Latency is a weaker claim and is presented as one. The previous project's own two runs of the identical System B recorded 3,993 ms and 7,058 ms — a 3,065 ms spread from provider variance alone. Against B+ the improvement is large and consistent with removing most of the model calls; against B it is inside the noise. The defensible statement is that this build does the same work in roughly one call instead of four, and latency followed.

The two accuracy columns are not directly comparable: the previous action accuracy was scored by an LLM judge over 51 cases against a different scope and a different corpus. This build's equivalent is 52/52 on deterministic routing, measured differently and reported separately below.

Evaluation, by dimension

Against the previous build's own evaluation — 66 responses per model scored objectively, and a designed human review of 132 responses across four dimensions with four reviewers.

	METRIC	PREVIOUS	THIS BUILD
Safeguarded	safeguarding recall	1.000	1.000 · 12/12
	safeguarding precision	0.800	1.000 · 12/12
	invented contact details	0 / 132	0
Accurate	grounded answers with no citation	0 / 39	0 shipped †
	mean citations per answer	2.46	2.61
	routing accuracy	0.766 F1	52/52
	retrieval Adequate@5	–	0.960
Reliable	**unusable outcomes**	12/51 · 23.5%	3/52 · 5.8% ‡
	latency p50 / p95	14.2 s / 22.8 s	5.2 s / 14.8 s
	errors	0	0
	forbidden retrievals	–	0/23
Resonant	tone and inclusion	4.61 / 5	*reviewer scored*
	Kenyan register	4.00 / 5	*reviewer scored*
	conversation	4.06 / 5	*reviewer scored*
	register match, measured	–	38/38
	natural continuation	–	32/37 · 86%
	deferrals avoided	–	38/38
Cost	LLM calls per turn	3.53	0.69
	tokens per turn	6,838	4,946
	cost per turn	$0.0189	$0.0109

† One draft was blocked by the validator for having no citation, so nothing uncited reached a girl. ‡ 1 validator block, 2 no-evidence, 0 provider errors; correct out-of-scope declines excluded. Where an instrument differs between the builds, the row is directional rather than exact.

The resonance rows are the honest gap

The previous build ran a designed human review: 132 responses, two models shown **side by side and unlabelled** so a preference could not be a preference for a brand, four reviewers, and one sheet per dimension because a reviewer asked to judge everything at once judges tone and calls it quality. Disagreement was kept rather than averaged away.

Those scores are people's judgement, and this build has no equivalent. Its resonance rows are deterministic proxies: they measure whether a property holds, not whether a reply is good. **Reproducing that review is the single most valuable next step**, and it is what would turn 86% continuity from an engineering number into a product one.

Why the unusable rate is the number to look at

The previous configuration recorded **zero unsafe actions alongside a 23.5% unusable rate**. Excellent on safety, and a poor product, because a system that refuses everything is perfectly safe. That build's own review put it plainly:

> "Service hallucination protection works. Service fulfilment does not." — *reviewer*

Zero invented numbers across 132 responses, and it still would not hand over a helpline when asked for one. This build attaches verified contacts on every route that should carry them.

> **Safety is not usefulness**, and measuring only one of them is how you ship the other.

The previous build's own component analysis isolates the cost precisely, because it built System B+ specifically to separate the two additions:

STEP	ADDS	COST IN UNHELPFUL REFUSALS	UNSAFE PREVENTED
B → B+	output validator	+1	–
B+ → C	LLM evidence judge	+6	1

The output validator is effectively free; the judge costs six times that for one prevented case. So this build keeps deterministic output validation and drops the judge — which is that project's own recommendation, not a new one.

Architecture

`validate → safeguarding screen → route → resolve* → prepare* → retrieve → generate → check`

** conditional. Neither is a universal stage.*

Only generation uses a hosted LLM. Routing, safeguarding, resolution, query preparation and validation are deterministic rules. Embeddings run locally on `bge-m3`, so no question leaves the machine to be encoded and there is no per-query embedding cost.

The safeguarding screen is a gate, not a route. It runs first on every message, and when it fires nothing downstream executes: no retrieval, no model call, no validator. That ordering is the single most important decision in the build, because retrieval cannot decline — a deliberately out-of-scope question was measured retrieving at 0.668, above most in-scope ones.

Two output contracts, chosen by provenance

	GROUNDED	CONVERSATIONAL
Used for	factual, access	greetings, support, her ambitions
Safe because	every claim carries a citation	it makes no claim at all
Blocked on	an uncited claim	a citation marker with no passage

Which contract applies is decided by which path produced the turn, never inferred from whether citations happen to be present.

Detect broadly, escalate narrowly

One safeguarding route, two severities, implemented as one boolean. **Urgent** — force, threat, assault, contraceptive sabotage, self-harm — puts verified contacts in front of her. A **concern** — pressure, conditional consent, *"he keeps asking"* — is recognised as safeguarding and answered as a conversation, with contacts offered rather than pushed. Treating every coercion signal as an emergency both reads as being passed on when she came to talk, and at scale buries the services in cases that were never emergencies. Nothing contacts anyone on her behalf.

Evaluation

Five evaluations over 121 labelled items, each with criteria fixed before the run.

1 · Decision layer — 52 messages

METRIC	RESULT	NOTE
Overall accuracy	52/52	1.000
Safeguarding recall	12/12	1.000
Safeguarding precision	12/12	1.000
Contrast pairs	6/6	same surface form, different path

2 · Retrieval — 31 questions, tagged by Theory-of-Change driver

METRIC	NATURAL	PREPARED	ORACLE
Adequate@5 — a passage that *answers* her	0.880	0.960	1.000
Hit@5	0.926	0.963	0.926
MRR	0.883	0.920	0.864
Agency drivers, mean MRR	0.611	0.750	0.889

Gold labels are source-level, so Hit@5 is an optimistic upper bound and is reported as one. Questions are tagged by the eight behavioural drivers rather than generic RAG categories, because knowledge is one driver of eight: this retriever scored knowledge MRR 1.000 while self-identity sat at 0.417.

3 · Multi-turn — 4 journeys, 23 turns

Forbidden passages **1 → 0**. Before conversation state existed, *"and does it hurt?"* asked after a question about the implant retrieved **female sterilization**, and *"where can I go?"* after a coercion disclosure retrieved **BTL**, which is permanent. Path accuracy 21/23; the two disagreements are disputed labels, ours, left uncorrected and documented.

4 · Mixed-intention messages — 15 messages

Wanted evidence **8 → 9** of 14, unwanted passages **4 → 3**, and a purely emotional control that now retrieves nothing at all. Clause-level retrieval, with no word list deciding which clause is the health question — the similarity scores do that.

5 · Conversation quality — 19 journeys, 38 turns

PROPERTY	RESULT
Grounded turns that cite a source	24/24 · 100%
Reply matches her register	38/38 · 100%
Usable reply, not a refusal	38/38 · 100%
Free of a deferral	38/38 · 100%
Free of machinery talk	37/38 · 97%
Free of a passive standing offer	36/37 · 97%
Gives her somewhere to go next	32/37 · 86%

There is no LLM judge, and warmth is deliberately not measured. Every property above is defined in advance and checkable by reading the text, so the score means the same thing on every run and any row can be verified by hand. Warmth cannot be scored honestly by a machine — a judge asked to try produces false positives on exactly the warm, personal turns that matter most — and naming that boundary is more useful than a number nobody can reproduce.

Two journeys, end to end

Girl Effect's Theory of Change runs **behavioural drivers → intent → service access → behaviour change**. Knowledge is one driver of eight, so a system can answer every question correctly and never move a girl toward a service. Two conversations are scored against it on every run, through the real pipeline.

A girl worried after unprotected sex, reaching HIV testing

SHE WRITES	DRIVER	STAGE REACHED	EVIDENCE
"niaje aunti"	rapport	rapport	–
"my friend says you can tell if someone has HIV by looking at them"	knowledge	knowledge	2 cited
"we didnt use protection last month and i keep thinking about it"	perceived risk	knowledge	–
"i should get tested but i am scared what they will say about me"	self-efficacy	confidence	–
"tell me what actually happens when you go for an HIV test"	**intent**	knowledge	3 cited + contacts
"where can i actually go for one?"	**service access**	service access	+ contacts
"asante sana aunti, i will text them tomorrow"	behaviour change	rapport	–

> "That fear is real, and it's probably the biggest thing standing between you and getting tested, not the testing itself. A lot of girls worry more about being judged than about the result. What are you imagining they'll say?"

The conversion turn returns the full testing process, cited, and ends with **Marie Stopes 0800 720 005** and **One2One 1190** — both text-capable, which matters because a girl on a shared phone often cannot make a call.

A girl held back by a myth and by family norms, reaching family planning

Answered in the register she wrote in:

> **She:** "watu wanasema family planning inaharibu mji wa mtoto. is that true?"
>
> **Aunti:** "Hiyo si kweli, na ni one of those misconceptions zinazoenea sana. Family planning haiharibu mji wa mtoto (uterus) au uwezo wa kupata mtoto baadaye. Njia nyingi za contraception ni reversible, meaning fertility inarudi haraka baada ya kuacha kutumia."

The barrier in that journey is social rather than informational — *"my mum would kill me if she found out i was even asking"* — and the turn that unlocks it is a rights question in her own words, *"can i even get family planning at my age without my parents agreeing?"*, which is the measured vocabulary gap the corpus states as *informed consent for adolescents and youth*.

ACROSS BOTH JOURNEYS, 14 TURNS	RESULT
stage reached as intended	14/14
citation where a fact was claimed	14/14
contact present exactly where intended	14/14
refusals	0/14

An access question never ends in a refusal. The answer to "where can I go" lives in the verified table rather than in generated text, so the table can answer it alone whatever happens to the draft. That turn is the Theory of Change's terminus and it is a guarantee rather than a probability.

Getting her to a service

The Theory of Change runs behavioural drivers → intent → **service access** → behaviour change. A system that answers every question well and never gets her to a service has done the easy half.

HER TURN	WHAT SHE GETS
"Where can I get family planning near me?"	the cited answer, plus Marie Stopes and One2One with numbers
"Where can I go for an HIV test?"	routed to the HIV/STI rows, off her words
discloses coercion, then "where can I go?"	the help pathway, **0 model calls**
self-harm risk	crisis lines in front of her, not behind a tap
pressure without force	contacts offered behind the tap, not pushed

Contacts are read from a verified table and **never generated**. No corpus chunk contains a phone number, so a number in generated text came from the model's memory, and the validator treats that as fatal. A row reaches a girl only once it carries a source, a checker and a date.

Scope and limitations

- **The evaluation sets are authored in-house** — 121 labelled messages and questions across four sets, written alongside the system. A rigorous regression harness and a design instrument; testing with real users is the next step.
- **The corpus is weighted toward provider guidance** — 1,326 clinical chunks against 58 written for a young reader, which is what clause-level retrieval and the vocabulary mappings exist to bridge.
- **Asking in Kiswahili costs 0.062 similarity**, measured over five matched pairs. Direct translation is handled; idiomatic Sheng is harder.
- **Continuity sits at 86%**, the newest metric and the one with most headroom.

What was deliberately not built

No LLM evidence judge, no output judge, no turn planner, no model-based query rewriter as a first resort, no conversation summariser, no entity tracker, no per-girl profile, no external monitoring vendor. Observability is a JSONL file holding operational fields only — no message text, no identifier for her — because an event log for a safeguarding product, kept by default, is a surveillance database with a dashboard on it.